

**SIMATS ENGINEERING**  
**CO3-ASSESSMENT TOOL 2**  
**Case study**

**COURSE NAME: Computer Networks**

**COURSE CODE:CSA0706**

**COURSE OUTCOME COVERED**

**CO3:** *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40% Dr.V.Parthipan

**QUESTIONS:4**

**Total Marks:40**

***Code of Conduct:***

*I D.Uday kiran (192425161) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

D.Uday kiran

Faculty Signature

## **CASE STUDY 1 – Cloud-Based Video Conferencing**

**Q1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.**

### **Answer**

#### Understanding of the Case

The company is using a cloud-based video conferencing system for daily client meetings. During peak hours, users experience voice distortion, frozen video, and delayed communication. The network has 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization.

These conditions indicate that the network is unable to provide the stable, low-latency communication required by real-time applications.

#### Application of Concepts

Video conferencing mainly requires low latency, low jitter, and continuous packet delivery. Therefore, UDP is generally more suitable than TCP for real-time audio and video.

TCP provides reliable and ordered delivery. When a packet is lost, TCP retransmits it, which can introduce additional delay. In video conferencing, receiving an old packet late may be less useful than continuing with the latest video data.

UDP does not retransmit lost packets and therefore avoids this additional delay.

#### Analysis

The 8% packet loss can cause missing audio and video information. The 180 ms jitter causes packets to arrive at irregular times, producing voice distortion and video freezing.

The fluctuating bandwidth can also cause congestion. If TCP is used for real-time media, retransmissions and congestion control may further increase delay.

However, the application layer also contributes if the jitter buffer is poorly configured. A very small buffer may cause packet loss, while an excessively large buffer increases delay.

Therefore, the primary issue is the transport protocol and network condition, with application-level buffering acting as a secondary factor.

**Q2. Justify your answer based on the communication requirements of real-time multimedia applications.**

### **Answer**

#### Understanding of the Case

Real-time multimedia applications such as video conferencing, VoIP, online classes, and live meetings require information to be delivered continuously and quickly.

In this case, users are experiencing delayed communication, frozen video, and voice distortion because the network conditions are not suitable for real-time traffic.

### Application of Concepts

The important requirements of real-time multimedia are:

- Low latency
- Low jitter
- Low packet loss
- Continuous data transmission
- Fast delivery of packets

UDP is suitable because it has low overhead and does not require acknowledgments or retransmission for every lost packet.

TCP, in contrast, guarantees reliable and ordered delivery. When a packet is lost, TCP retransmits it. This reliability can result in additional waiting time.

### Analysis

Consider a video conference where one video packet is lost.

With TCP, the lost packet is retransmitted. The receiver may have to wait for the missing data, increasing delay and potentially causing video freezing.

With UDP, the lost packet can simply be discarded and the receiver can continue processing newer packets. This is more suitable for live communication because a late packet is often less useful than a new packet.

However, UDP itself does not guarantee reliability. Therefore, the application should use techniques such as jitter buffering, packet-loss concealment, and adaptive bitrate control.

### Solution Design

The recommended approach is:

UDP → RTP → Adaptive Jitter Buffer → Audio/Video Decoder

The system can dynamically adjust the video quality according to available bandwidth.

## **Q3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.**

### **Answer**

#### Understanding of the Case

The major problems are 8% packet loss, 180 ms jitter, fluctuating bandwidth, voice distortion, and frozen video. Therefore, both network-level and application-level improvements are required.

#### Application of Concepts

At the protocol level, real-time communication can use UDP with RTP. QoS mechanisms can prioritize voice and video traffic over less time-sensitive traffic.

At the application level, the conferencing system should dynamically respond to changes in network conditions.

## Analysis

The following improvements can reduce the problems:

### 1. Use UDP/RTP

UDP avoids retransmission delays, while RTP provides sequence numbers and timestamps for real-time media.

### 2. Implement QoS

Voice and video packets should receive higher priority so that they experience less delay and jitter during peak traffic.

### 3. Adaptive Jitter Buffer

The application should dynamically adjust its jitter buffer according to network conditions.

### 4. Adaptive Bitrate

When bandwidth decreases, the application should automatically reduce the video bitrate.

### 5. Dynamic Video Resolution

The system can switch from high-definition video to lower resolution when network conditions deteriorate.

### 6. Forward Error Correction

FEC can add redundant information that helps recover some lost packets without requiring retransmission.

### 7. Packet-Loss Concealment

For audio, missing packets can be estimated or concealed to reduce noticeable voice distortion.

### 8. Network Capacity Improvement

Additional bandwidth and better traffic management should be provided during peak business hours.

## Solution Design

A suitable solution can be represented as:

UDP/RTP + QoS + Adaptive Bitrate + Jitter Buffer + FEC + Packet-Loss Concealment

This combination allows the conferencing application to continue operating even when network conditions change.

## Expected Outcome

The proposed solution should:

- Reduce voice distortion
- Reduce video freezing
- Reduce communication delay
- Handle packet loss more effectively
- Reduce the effect of jitter
- Improve overall conferencing quality

## CASE STUDY 2 – Global DNS Infrastructure

### Q1. Analyze the reasons behind the high DNS resolution time.

#### Understanding of the Case

The multinational software company hosts its web services across different continents. Employees are experiencing long delays before the application homepage loads. The average DNS lookup time is approximately 800 ms, which is very high for a web application.

DNS is responsible for converting a domain name into its corresponding IP address. The browser normally needs this IP address before it can establish communication with the web server.

#### Application of Concepts

Several factors can contribute to high DNS resolution time:

- Centralized DNS server: If requests are handled by a limited number of servers, the server can become overloaded.
- Geographical distance: Users located far from the DNS server may experience higher round-trip time.
- Network congestion: Heavy traffic can delay DNS queries and responses.
- Multiple DNS lookups: Recursive resolution may require communication with several DNS servers.
- Poor DNS caching: If DNS records are not cached effectively, repeated queries are sent to DNS servers.
- Server overload: A large number of simultaneous DNS requests can increase processing time.
- DNS server failure: Failed requests can cause retries and additional delays.

#### Analysis

An 800 ms DNS lookup means users have to wait nearly 0.8 seconds before the actual application connection can begin. For a global organization with users distributed across continents, sending all DNS queries to a distant or overloaded server can significantly

increase this delay.

For example, if an employee in India must contact a DNS server located in North America, the query has to travel a long geographical distance. During peak traffic, congestion and server load can further increase the response time.

Therefore, the major causes are centralization, geographical distance, network congestion, poor caching, and DNS server overload.

**Q2. Propose a suitable DNS architecture using Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.**

Understanding of the Case

The target is to reduce DNS resolution time from approximately 800 ms to below 50 ms while maintaining continuous service availability.

Application of Concepts

A distributed Anycast DNS architecture is suitable for this requirement.

The proposed architecture is:

Global Users

↓

Anycast DNS Address

↓

Nearest Regional DNS Server

↓

DNS Cache

↓

Authoritative DNS Servers

Multiple DNS servers should be deployed in different geographical regions.

For example:

- Asia
- Europe
- North America
- Australia

Anycast DNS

Anycast allows multiple DNS servers in different locations to use the same IP address. Routing mechanisms direct a user's DNS request to an appropriate nearby server.

This reduces the physical distance between the user and DNS server, thereby reducing latency.

## DNS Pre-fetching

DNS pre-fetching allows frequently required domain names to be resolved before the user actually requests them.

For example, if the homepage frequently loads resources from `api.example.com`, the DNS information for that domain can be obtained in advance.

This reduces waiting time during application loading.

## TTL Optimization

Time To Live (TTL) determines how long DNS records can remain in a cache.

For stable records, a suitable longer TTL can be used. This reduces the number of repeated DNS queries.

For frequently changing records, a shorter TTL should be used so that changes are reflected more quickly.

## Solution Design

The complete solution can be represented as:

User → Nearest Anycast DNS → Local Cache → Authoritative DNS

If one regional DNS server fails, routing can direct requests to another available DNS server.

## Expected Outcome

The proposed architecture should:

- Reduce geographical latency
- Reduce DNS server load
- Increase DNS cache hits
- Provide redundancy
- Improve global availability
- Help reduce lookup time toward the below-50-ms target

**Q3. Evaluate the advantages and limitations of the proposed architecture in terms of performance, scalability, and availability.**

## Performance

### Advantages:

Anycast directs users toward an appropriate nearby DNS server, reducing network round-trip time. DNS caching further reduces the number of queries that need to reach authoritative servers.

DNS pre-fetching can also reduce the waiting time experienced by users during application loading.

Limitation:

If a regional DNS server becomes overloaded or the routing path is inefficient, users may still experience increased latency.

Scalability

Advantages:

The distributed architecture can handle a much larger number of DNS queries because requests are distributed across multiple DNS servers.

As the number of users increases, additional DNS servers can be deployed in new regions.

Limitation:

Deploying and maintaining DNS servers across multiple geographical locations increases infrastructure and management complexity.

Availability

Advantages:

Multiple DNS servers provide redundancy. If one server or region fails, users can be directed to another available DNS server.

Anycast also provides automatic traffic distribution toward available locations.

Limitation:

DNS configuration errors or incorrect routing advertisements can affect multiple regions. Therefore, monitoring and proper configuration are essential.

Comparison

| Factor       | Advantage                    | Limitation                                |
|--------------|------------------------------|-------------------------------------------|
| Performance  | Lower DNS latency            | Routing problems can affect performance   |
| Scalability  | Handles large global traffic | Higher infrastructure complexity          |
| Availability | Provides server redundancy   | Configuration failures can affect service |
| Caching      | Reduces repeated DNS queries | Stale records may remain cached           |



| Factor  | Advantage                               | Limitation                          |
|---------|-----------------------------------------|-------------------------------------|
| Anycast | Directs users to suitable DNS locations | Requires careful routing management |
| TTL     | Controls caching efficiency             | Incorrect TTL can delay DNS updates |

### CASE STUDY 3 – Distributed DNS for Global Cloud Services

#### Q1. Examine the impact of centralized DNS architecture on application performance.

##### Understanding of the Case

The cloud service provider delivers applications to millions of users worldwide. During promotional events, a large number of users access the services simultaneously. The investigation shows that DNS queries take nearly 800 ms because all requests are directed to a single primary DNS server.

DNS is an important part of application access because the client must obtain the IP address of the cloud service before establishing communication with the application server.

##### Application of Concepts

A centralized DNS architecture means that DNS requests from users in different geographical locations are handled by one primary DNS server.

When millions of users send requests to the same server, it can become a bottleneck. The server has to process a very large number of queries, resulting in increased response time.

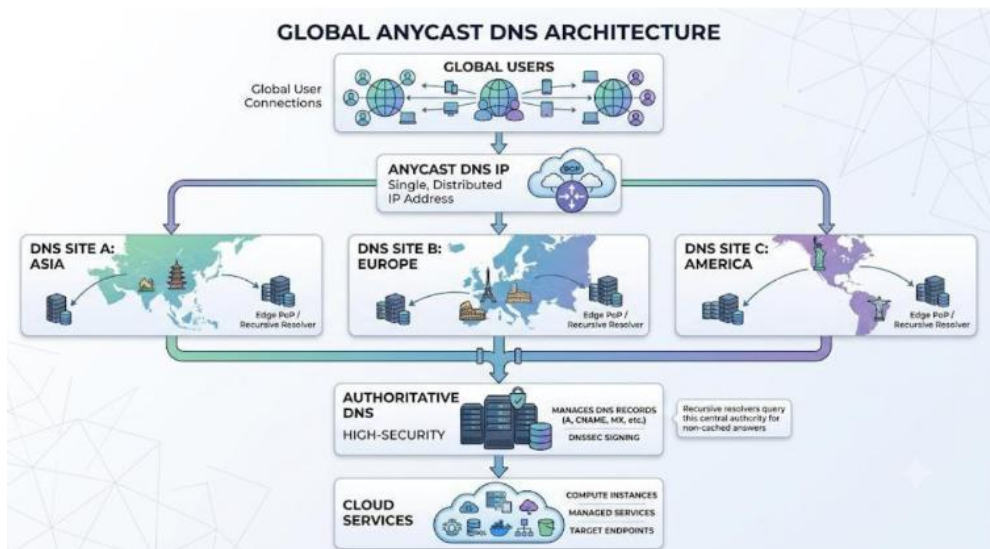
##### Analysis

The centralized architecture causes several performance problems:

- High DNS latency: Users far from the DNS server experience longer round-trip times.
- Server overload: Promotional events generate a large number of simultaneous DNS queries.
- Network congestion: Large numbers of DNS requests can increase network traffic.
- Single point of failure: If the primary DNS server fails, users may not be able to resolve the application domain.
- Increased application startup time: The application cannot establish communication until DNS resolution is completed.
- Poor scalability: A single DNS server cannot efficiently handle rapidly increasing global traffic.

The 800 ms DNS lookup time is particularly harmful because it adds significant delay before the application connection begins.

## Solution Design



**Q2. Design a**

**distributed DNS solution that minimizes lookup latency while maintaining high availability.**

### Understanding of the Case

The organization requires a DNS system capable of handling millions of global requests while minimizing DNS lookup latency and maintaining continuous availability.

### Application of Concepts

A distributed DNS architecture can use Anycast routing, regional DNS servers, DNS caching, and redundant authoritative servers.

The proposed architecture is:

Global Users



Anycast DNS IP



Nearest/Optimal DNS Resolver



Regional DNS Cache



Multiple Authoritative DNS Servers

### Analysis

Multiple DNS servers should be deployed across major geographical regions such as:

- Asia
- Europe
- North America
- Australia

All servers can provide DNS service using an Anycast IP address.

When a user sends a DNS request, network routing directs the request toward an appropriate DNS server. A nearby DNS server can respond much faster than a single centralized server located far away.

### Anycast Routing

Anycast allows multiple DNS servers in different locations to advertise the same IP address.

Its main benefits are:

- Lower latency
- Traffic distribution
- Geographic proximity
- Better fault tolerance
- Automatic redirection during failures

### DNS Caching

Frequently requested DNS records should be cached. If the required record is already available in the cache, the DNS server can respond without contacting the authoritative server.

This significantly reduces lookup time and authoritative-server load.

### Redundancy

Multiple authoritative DNS servers should be deployed. If one server becomes unavailable, another server can continue providing DNS responses.

### Solution Design

The final architecture is:

Users → Anycast DNS → Regional DNS Server → Cache → Authoritative DNS

If one regional server fails:

User → Anycast DNS → Another Available DNS Server

Expected Outcome

This design should:

- Reduce DNS lookup latency
- Handle large numbers of queries
- Avoid a single point of failure
- Improve global availability
- Distribute DNS traffic
- Improve scalability during promotional events

### **Q3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.**

#### Understanding of the Case

The cloud service provider needs to handle millions of DNS requests efficiently while maintaining high availability. TTL, Anycast, and caching play important roles in achieving this goal.

#### Application of Concepts

##### 1. TTL Values

Time To Live (TTL) determines how long a DNS record can remain in a resolver's cache.

A suitable TTL reduces repeated DNS queries.

For example, if a DNS record has an appropriate TTL, thousands of users can use the cached record without contacting the authoritative DNS server repeatedly.

#### Analysis of TTL

##### Higher TTL:

- Reduces DNS query frequency
- Reduces server load
- Improves response time
- Improves scalability

However, very high TTL values can cause outdated DNS records to remain in caches for a long time.

##### Lower TTL:

- Allows DNS changes to propagate faster

- Provides more control over changing records

However, it increases the number of DNS queries and server load.

Therefore, an optimized TTL should be selected based on how frequently DNS records change.

## 2. Anycast Routing

Anycast allows multiple DNS servers to use the same IP address.

A user's DNS request is routed toward an appropriate DNS server based on network routing.

Advantages:

- Reduces geographical latency
- Distributes DNS traffic
- Improves availability
- Provides automatic failover
- Prevents a single-server bottleneck

If one Anycast location becomes unavailable, routing can direct requests toward another available location.

## 3. DNS Caching

DNS caching stores previously resolved DNS records.

For example, if 10,000 users request the same domain, the DNS resolver can return the cached IP address rather than contacting the authoritative DNS server 10,000 times.

This provides:

- Faster DNS responses
- Lower network traffic
- Reduced authoritative server load
- Better scalability

## Solution Design

The three techniques work together:

User → Anycast DNS → Regional Cache → Cached Response

If the record is not available in the cache:

Regional DNS → Authoritative DNS → Cache Response → User

## Expected Outcome

| Technique        | Performance Benefit        | Fault-Tolerance Benefit                     |
|------------------|----------------------------|---------------------------------------------|
| TTL Optimization | Reduces repeated queries   | Helps control DNS update behavior           |
| Anycast          | Reduces geographic latency | Provides routing-based failover             |
| DNS Caching      | Provides faster responses  | Reduces dependency on authoritative servers |

## CASE STUDY 4 – TCP Latency in Electronic Stock Trading

**Q1. Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.**

Understanding of the Case:

The financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms.

Network engineers have observed:

- Increased TCP retransmissions
- Longer connection establishment time
- Excessive buffering
- Increased transaction latency

The three major TCP-related factors are flow control, Nagle's algorithm, and SYN queue limitations.

### Application of Concepts

#### 1. TCP Flow Control

TCP uses a receive window to control the amount of data that can be sent before receiving acknowledgments.

If the receiver's buffer becomes full or the receive window becomes small, the sender must slow down. This can result in additional waiting time and increased transaction latency.

#### 2. Nagle's Algorithm

Nagle's algorithm combines small TCP segments into larger segments to improve network efficiency.

Although this reduces the number of small packets, it can introduce a delay because TCP may wait for additional data before transmitting a small message.

In a stock trading system, buy and sell orders are usually small but highly time-sensitive. Therefore, this delay can negatively affect order processing.

### 3. SYN Queue Limitations

TCP connection establishment uses the three-way handshake:

SYN → SYN-ACK → ACK

During market opening hours, thousands of connection requests may arrive simultaneously.

If the server's SYN queue becomes full, new connection requests may be delayed or dropped. This results in retransmissions and longer connection establishment times.

#### Analysis

These three factors can explain the increase in latency from 1 ms to 40 ms:

| Factor               | Effect                                                  |
|----------------------|---------------------------------------------------------|
| Flow Control         | Limits data transmission when receiver buffers are full |
| Nagle's Algorithm    | Can delay small trading messages                        |
| SYN Queue Limitation | Delays new TCP connections                              |
| Retransmissions      | Increase communication delay                            |
| Excessive Buffering  | Creates additional waiting time                         |

## Q2. Explain how each factor affects transaction processing in high-frequency systems.

### Understanding of the Case

High-frequency trading systems process a large number of transactions within very short time intervals. Even a few milliseconds of additional delay can affect transaction processing and order acknowledgment.

### Application of Concepts

#### 1. Effect of Flow Control

TCP flow control prevents the sender from overwhelming the receiver.

If the receiver's buffer is full, TCP reduces the transmission rate. In a high-frequency trading system, this can create a queue of pending transactions.

As a result, orders may take longer to reach the server or acknowledgments may take longer to return.

## 2. Effect of Nagle's Algorithm

Nagle's algorithm is designed to reduce network overhead by combining small packets.

However, trading applications often send small, independent messages such as:

- Buy orders
- Sell orders
- Price updates
- Order acknowledgments

Waiting to combine these small messages can add unnecessary latency.

For high-frequency trading, even a small delay can be significant.

## 3. Effect of SYN Queue Limitations

When many clients establish connections simultaneously, the server must process many SYN requests.

If the SYN queue becomes overloaded:

- Connection requests may wait.
- SYN packets may be dropped.
- Clients may retransmit SYN packets.
- Connection establishment takes longer.

This directly increases the time required to process new trading connections.

## Analysis

In a high-frequency environment, these delays accumulate. A transaction that normally takes 1 ms may take 40 ms during peak periods because of buffering, retransmissions, connection delays, and TCP processing.

## Impact on Trading

Increased latency can result in:

- Delayed order acknowledgments
- Delayed trade confirmations
- Missed trading opportunities
- Increased customer dissatisfaction
- Increased network retransmissions



- Reduced system efficiency

### **Q3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.**

#### Understanding of the Case

The trading platform requires very low latency and high reliability. The current latency has increased from 1 ms to 40 ms, so TCP and server configurations need to be optimized.

#### Application of Concepts

The following TCP configuration changes are recommended.

##### 1. Disable Nagle's Algorithm

The application can use:

`TCP_NODELAY`

This allows small trading messages to be transmitted immediately rather than waiting to combine them with other data.

This is particularly useful for small and time-sensitive trading messages.

##### 2. Increase SYN Queue Capacity

The server's SYN queue should be configured to handle large numbers of simultaneous connection requests.

This reduces the possibility of dropped SYN requests and unnecessary retransmissions during market opening.

##### 3. Optimize TCP Receive and Send Buffers

TCP buffer sizes should be configured according to the application's traffic requirements.

Proper buffer configuration prevents unnecessary flow-control limitations while avoiding excessive buffering.

##### 4. Use Persistent TCP Connections

Where appropriate, the system should maintain existing TCP connections instead of establishing a new connection for every order.

This avoids repeated three-way handshakes and reduces connection establishment latency.

##### 5. Reduce Packet Loss

Network packet loss should be minimized because lost TCP packets cause retransmissions.

Improving network quality and reducing congestion can significantly reduce TCP retransmission delays.

## 6. Reduce Excessive Buffering

The system should avoid unnecessarily large queues and buffers. Excessive buffering can increase waiting time and create additional latency.

## 7. Monitor TCP Performance

The organization should continuously monitor:

- TCP retransmissions
- RTT
- Connection establishment time
- Receive window
- Send window
- SYN queue usage
- Packet loss

This helps identify the exact source of latency during peak trading periods.

## Solution Design

A suitable optimized configuration is:

Persistent TCP Connections + TCP\_NODELAY + Optimized Buffers + Increased SYN Queue + Reduced Packet Loss + Continuous Monitoring

## Expected Outcome

The proposed changes can help:

- Reduce order acknowledgment latency
- Reduce TCP retransmissions
- Improve connection establishment
- Reduce buffering delays.